Statistical analysis plan (SAP) for the TenCRAOS trial

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STATISTICAL ANALYSIS PLAN for the TenCRAOS trial

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STATISTICAL ANALYSIS PLAN for the TenCRAOS trial

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ABBREVIATIONS

AE	Adverse event
CI	Confidence interval
DMC	Data monitoring committee
EAE	Expected adverse events
eCRF	electronic case report form
EDC	Electronic data capture
FAS	Full analysis set
IMP	Investigational medicinal product
MedDRA	Medical dictionary for regulatory activities
SAE	Serious adverse event
SAP	Statistical analysis plan
SAS	Supporting analysis set
SD	Standard deviation
SOC	System organ class
SUSAR	Suspected unexpected serious adverse reaction
TEAE	Treatment emerging adverse event
TESAE	Treatment emerging serious adverse event

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STATISTICAL ANALYSIS PLAN for the TenCRAOS trial

1 Introduction

This SAP covers the statistical analysis of the TenCRAOS trial (safety and efficacy analysis of the primary and secondary endpoints) included in the primary and follow-up publications. Primary publication will include data from the acute phase of the trial and follow-up until day 30 (±5 days)*. 90-day follow-up will be completed in June 2025, and presented in a separate publication.* See protocol for a more extended introduction to the topic. The SAP is developed with guidance on the required content of statistical analysis plans (SAPs) to support transparency and reproducibility[1].

1.1 Study rationale

Central retinal artery occlusion (CRAO) is an ophthalmologic emergency that, without prompt revascularisation, bears high risk of permanent blindness. The condition is typically the result of an artery- to-artery embolism from a carotid plaque or cardioembolism. Different treatment options have been tested, such as sublingual isosorbide dinitrate, topical timolol maleate, systemic pentoxifylline, inhalation of 10% carbon dioxide, hyperbaric oxygen, ocular massage, intravenous acetazolamide, mannitol, anterior chamber paracentesis, hemodilution, and corticosteroids. However, none have been shown to be more effective than placebo. Tenecteplase is a tissue plasminogen activator (tPA) produced by recombinant DNA technology. It is a well-established revascularising therapy in acute myocardial infarction and has also been used in acute ischemic stroke. Rapid administration of systemic thrombolytic agents has proved to be an effective treatment for acute ischemic stroke within 4.5 hours of symptom onset and is a promising treatment for CRAO, but there has been no RCT comparing early systemic thrombolysis to placebo in CRAO. The aim of this project is to assess the efficacy and safety of systemic tenecteplase within 4.5 hours onset of CRAO in a large multi-site trial. The hypothesis that will be tested in this study is that intravenous tenecteplase + placebo within 4.5 hours of CRAO onset improves visual acuity after 30 (5) days compared to placebo + acetylsalicylic acid.

1.2 Trial Objectives

1.2.1 Primary Objective

To assess whether treatment with systemic Tenecteplase + placebo within 4.5 hours onset of CRAO increases the proportion of patients reaching an improvement of at least 0.3 logMAR best-corrected visual acuity (≤ 0.7 logMAR) in the affected eye at 30 (±5) days after treatment in comparison to patients treated with placebo + acetylsalicylic acid.

1.2.2 Primary Endpoint

Primary endpoint is the poportion of patients with ≤ 0.7 logMAR visual acuity in the affected eye at 30 (±5) days after treatment. The primary endpoint variable gets values 0 (logMAR best-corrected visual acuity >0.7) or 1 (logMAR best-corrected visual acuity ≤ 0.7).

2 Trial Methods

2.1 Trial Design

- Randomized trial
- Controlled
- Double-dummy
- Double-blind
- Parallel group
- Multi-centre
- Study intervention assignment by 1:1 central computer randomization procedure stratified by investigational center
- Treatment groups: TNK 0.25mg/kg + placebo (active), or ASA + placebo (placebo)
- Adult patients with non-arteritic central retinal artery occlusion with ≥ 1.0 logMAR visual acuity and symptoms lasting less than 4.5 hours
- Treatment administered within 4.5h of symptom onset
- The primary endpoint is logMAR best-corrected visual acuity ≤ 0.7 at 30 (±5) days after treatment (yes/no)
- Duration of study intervention: tenecteplase/placebo as a single intravenous bolus over approximately 10 seconds + capsules with ASA/placebo, time before primary outcome 30 days, and total study participation 90 days.

2.2 Randomization

Eligible patients will be allocated in a 1:1 ratio between tenecteplase and placebo, using a blocked computer randomization procedure. Details of block size and allocation sequence generation will be provided in a separate document unavailable to those who enroll patients or assign treatment.

Treatment kits containing either active treatment (tenecteplase) and inactive co-medication (placebo) or inactive treatment (placebo) and active co-medication (acetylsalicylic acid) will be packed by the pharmacy according to the computer-generated random allocation sequence. Kits will be shipped to each site in complete blocks. Each treatment kit will be given a unique kit number according to the allocation sequence. Only personnel authorised by the principal investigator for preparing treatment will have access to treatment allocation, and the allocation will not be available until the patient has signed the informed consent form and deemed eligible to participate in the study. Once the patient has been included, the site person authorised for treatment preparation will open and use the treatment kit with the lowest kit number.

2.3 Sample size

Sample size calculation has been based on the primary endpoint "proportion of patients with ≤ 0.7 logMAR visual acuity at 30 (±5) days, after treatment", representing an improvement in visual acuity of at least 0.3 logMAR. The proportion of patients reaching the primary endpoint (≤ 0.7 logMAR) is assumed to be 20% in the placebo group. Further, we assume that 50% of the patients receiving active treatment will reach the primary endpoint by 30 days. Assuming this risk difference of 30% between the treatment groups, a sample size of 78 patients (39 in each group) is expected to achieve a power of 80% to detect the difference on a 5% significance level ($p<0.05$, two-sided). Inclusion of new patients will continue until 78 patients have been randomised.

2.4 Statistical Framework

2.4.1 Hypothesis Test

This trial is designed to establish the superiority of systemic Tenecteplase + placebo within 4.5 hours onset of CRAO in comparison to placebo + acetylsalicylic acid in terms of proportion of patients reaching an improvement of at least 0.3 logMAR best-corrected visual acuity (≤ 0.7 logMAR) in the affected eye at 30 (±5) days after treatment. The primary endpoint variable is a binary (yes/no) visual acuity of at least logMAR 0.7 (< logMAR 0.7) at 30 days after treatment. The primary null hypothesis is that there is no difference in the proportion of patients with best-corrected visual acuity of < logMAR 0.7 between patients receiving tenecteplase + placebo in comparison to patients receiving placebo and acetylsalicylic acid. The primary null hypothesis is evaluated in the full analysis set (FAS; intention-to-treat approach).

2.5 Statistical Interim Analyses and Stopping Guidance

There will be one pre-planned interim safety analysis that will assess the safety of the interventions during the trial, with the intent to stop the study early if there is a safety concern. The interim safety analysis will be performed after 20 and 50 patients have been randomised. Based on the safety data presented in the DMC meeting, the DMC will give a recommendation for the trial steering committee either to continue the trial or to stop the trial due to a safety concern. After that, the steering committee decides whether they want to stop the trial or not. If the trial is stopped, then all new inclusions stop, but the patients already included will be followed-up according to the protocol. After the last-patient-last-visit, the standard procedure for the trial ending starts: data will be cleaned and monitored, SAP will be written, and the final analyses ran.

2.6 Timing of Final Analysis

The final analysis will be performed in a two-step set-up. First, the primary outcome analyses, secondary analyses, baseline characteristics and safety assessments including timepoints up to 30 days will be analyzed when all patients have concluded their 30-day follow-up, see Table 2.1, and Section 4.5.2. Second, the remaining analyses will be performed once all patients have concluded their 90-day follow up (June 2025).

The first analysis is planned when all patients have concluded their 30-day follow-up, all data have been entered, verified and validated and the database has been locked. Before the database lock, trial statistician will export mock datasets with a pseudo random allocation of patients to different treatment arms in order to initiate writing the code for the analyses and preparing a dummy results report. Statistician will also perform test analysis to detect any outliers or missing data so that the issues can be addressed in the SAP (blind review). A complete SAP including all analyses for the two-step analysis plan will be finalized and the SAP will be reviewed by another at the clinical trial unit, Oslo University Hospital for quality control before final approval. The database lock for all visits up to and including visit at day 30 will be performed after this SAP has been signed by the study group. After the database lock, the final datasets for the first round of analysis will be delivered to the statistician for the analysis together with the true randomization code, unblinding the trial for all unblinded personnel (including the study statistician).

The second round of analysis is planned when all patients have concluded their 90-day follow-up, all new data after follow-up at day 30 have been entered, verified and validated and the database has been locked from day 30 up to and including day 90. After the database lock, the final datasets for analysis predefined in the SAP described above will be delivered to the statistician for the second round of analysis.

Table 2.1. Time points and assessments included in the primary publication.

Study requirements	Baseline	Day 1, 24 (18-36) h after treatment	Discharge after 36h (recorded if discharge > 36h after treatment)	30 (±5) days/Early Withdrawal
Ophthalmologic assessment	X			X
Medical history	X			
Weight	X			
Physical examination[3]	X	X	X	X
NIHSS	X	X	X	X
Brain CT/MRI	X	X		
Blood pressure, pulse, temperature	X	X	X	X
Quality of life assessment (EuroQol 5D-5L)				X
National Eye Institute Visual Functioning Questionnaire 25 (NEI-VFQ 25)				X
Adverse events		X	X	X
Relevant concomitant medication	X	X	X	X
Modified Rankin Scale (mRS)	X	X	X	X
Recording of protocol deviations	X	X	X	X

2.7 Timing of Outcome Assessments

For all clinically planned measures, visits should occur within a window of the scheduled visit. The target day and visit window is defined in as:

Visit Label	Target Day	Definition (Day window)
Baseline	Day 0	Symptom onset, arrival at study centre, arrival at the stroke centre
Treatment allocation	Day 0	Randomisation/ treatment allocation within 4.5h from symptom onset
Day 1: 18-36h after treatment	Day 1	18-36h after treatment
Discharge after 36h	Discharge from hospital	Recorded only if patient was discharged >36h after treatment allocation
30 days	Day 30	30 days after treatment $\pm$ 5d
90 days	Day 90	90 days after treatment $\pm$ 15d

3 Statistical Principles

3.1 Confidence Intervals and p-values

All p-values are two-sided. If a p-value is below 0.05, the corresponding treatment group difference will be denoted as statistically significant. All efficacy estimates will be presented with two-sided 95% confidence intervals. As there is only one primary null hypothesis to be tested in this trial, there will be no adjustments for multiplicity. The secondary endpoint efficacy estimates will be presented with unadjusted 95% confidence limits.

3.2 Adherence and Protocol Deviations

3.2.1 Compliance to Allocated Treatment

The study treatment is administered as a single intravenous bolus over approximately 10 seconds, and as oral capsules with 300mg ASA/placebo at the discretion of the study site personnel. Only one cycle of treatment is administered, regardless of any disease progression. The dosage of tenecteplase is 0.25mg/kg with a maximum dose of 25mg. The dose, date and time of administered treatment will be recorded in the source documents and recorded in the CRF.

3.2.2 Protocol Deviations

The following are pre-defined protocol deviations regarded to affect the efficacy of the intervention:

1. Subjects entered the study even though they did not meet the entry criteria.
2. Subjects who developed withdrawal criteria (from treatment or study) during the study but were not withdrawn.
3. Subjects who received the wrong treatment or incorrect dose.
4. Subjects who received an excluded concomitant treatment.

5. Failure to collect data necessary to interpret the primary endpoint:
    - Visual acuity score at baseline and at 30 days
6. Failure to collect:
    - Brain CT/MRI at baseline and 24h
    - Visual acuity score at 90 days
    - Monocular Esterman perimetry score at 30/90 days
7. Hemorrhagic events after administration of the study medication, especially intracranial hemorrhages
8. Other Serious Breaches according to Guideline for the notification of serious breaches of Regulation (EU) No 536/2014
    - SAE notification / safety procedures
    - Issues with informed consent

Reported Protocol deviations are classified prior to the final analysis as important or not important. Any patient with important protocol deviation will be excluded from the per-protocol (PP) population. The final list of patients to be excluded from the PP will be approved by the study group. The number and % of patients with major protocol deviations will be summarized by treatment group with details of type of deviation provided. The patients that are included in the FAS will be used as the denominator to calculate the percentages. No formal statistical testing will be undertaken.

3.3 Analysis Populations

Intention-to-treat (ITT) population is defined as all randomised patients, regardless of protocol adherence.

The Full Analysis Set (FAS) is defined as all patients randomly assigned to a treatment group, meeting the major entry criteria, and having been administered the study treatment within 4.5h of the symptom onset. See below the list of randomized patients with deviations from the protocol relating to eligibility criteria who were decided to include/exclude from FAS based on the blinded review of the data:

In the blinded review of the data, the following deviations from the protocol were observed:

- DK-16-009: patient had a baseline visual acuity 0.8 logMAR, i.e., 0.2 higher than the inclusion threshold of 1.0 logMAR. Still, the visual acuity was poorer than 0.7 logMAR in the acute phase and accordingly, the visual acuity could or could not improve to 0.7 logMAR following treatment. Thus, this deviation from the inclusion criteria was not considered major, and patient will be included in the FAS.
- DK-27-002: patient had a stroke after discharge and was therefore unable to attend the 30-day follow-up visit leading to a missing visual acuity measurement in the primary outcome. However, patient attended the 90-day follow-up. Because we do not expect the vision in CRAO to further improve from 30 to 90 days, the primary outcome measurement for this patient will be imputed with 90-day logMAR best-corrected visual acuity, and sensitive analyses for the missing data provided, as described in 5.2.1.4. Patient will be included in FAS.
- NO-21-004: Patient had subtle findings of a combined arteriovenous occlusion (exclusion criterion) evident at ophthalmologic 30-day follow-up that was not described by

ophthalmologist at inclusion. It is known from the literature that CRAO may develop into a combined arteriovenous occlusion, particularily following reperfusion. This 30-day finding was not considered important, and patient will be included in FAS.

- NO-6-002: Patient had a cilioretinal artery supplying the macula (exclusion criterion), and the primary endpoint of logMAR best-corrected visual acuity ≤ 0.7 at 30 days was met before any treatment. Accordingly, it would not be meaningful to include the patient in the primary analysis, and the patient will be excluded from FAS.
- DK-40-003: Patient had CRAO symptoms at wakeup, and therefore the exact time of symptom onset was unknown. Patient was included under the impression that wake-up symptoms were possible to include in the trial, and the baseline examination and study drug administration were performed promptly after arrival to the hospital. Patient will be included in the FAS, but excluded from the PP.
- DK-27-004: Re-examination of the ophthamological assement resulted in dismissal of the CRAO diagnosis, with vaso-spasms considered instead. This was considered a major violation of the entry criteria, and patient will be excluded from FAS.

Per-protocol (PP) population is a subset of the FAS consisting of patients who sufficiently comply with the protocol. Criteria for inclusion in the PP population will be specified in the statistical analysis plan section 3.2.2, and the final criteria will be defined prior to database lock. The following patient had protocol deviation classified as important and will be excluded form the PP analysis population:

- DK-40-003: Patient had CRAO symptoms at wakeup, and therefore the exact time of symptom onset was unknown. Patient was included under the impression that wake-up symptoms were possible to include in the trial, and the baseline examination and study drug administration were performed promptly after arrival to the hospital. Patient will be included in the FAS, but excluded from the PP.

Safety population includes all subjects who have been administered study treatment. Subjects who withdraw from the study will be included in the safety analysis. A list of withdrawn subjects, preferably with the reasons for withdrawal, will be made.

4 Trial Population

4.1 Screening Data, Eligibility and Recruitment

The total number of screened patients and reasons for not entering the trial will be summarized and tabulated.

A CONSORT flow diagram will be used to summarize the number of patients who were:

- assessed for eligibility at screening
- eligible at screening
- ineligible at screening*
- eligible and randomized
- eligible but not randomized*

- received the randomized allocation
- did not receive the randomized allocation*
- lost to follow-up*
- discontinued the intervention*
- randomized and included in the primary analysis
- randomized and excluded from the primary analysis*

*Reasons will be provided.

4.2 Withdrawal/Follow-up

The status of eligible and randomized patients at interim analysis and at the end of the trial will be tabulated by treatment group according to

- completed intervention and assessments
- withdrew consent*
- lost to follow-up*

*Reasons will be provided

Time from randomization to withdrawal/lost to follow-up will be presented by treatment group with median, 25/75 percentiles, minimum, and maximum.

4.3 Baseline Patient Characteristics

The following patient demographics, baseline characteristics and prognostic factors will be summarized:

Patient demographics
• Age at enrolment, years, mean (SD) • Sex, n (%) • Weight, mean (SD) • Baseline NIHSS, mean (SD) • Baseline acute ischemic lesions on diffusion-weighted MRI or brain CT, n (%) • Baseline CRP, mean (SD) • Baseline mRS, n (%) • Smoking status, n (%) • Hypertension, n (%) • Cardiac valvular disease, n (%) • Heart failure, n (%) • Coronary artery disease, n (%) • Extracranial carotid artery disease, n (%) • Atrial fibrillation, n (%) • Diabetes mellitus, n (%) • Dyslipidemia, n (%) • Previous transient ischemic attack (TIA) , n (%) • Previous ischemic stroke, n (%) • Previous intracerebral hemorrhage, n (%)

Patient demographics and baseline characteristics will be summarized by randomized treatment arm in the FAS, using descriptive statistics (N, mean/median, standard deviation/IQR) for continuous variables, and number and proportion of patients for categorical variables. Descriptive statistics will be based on non-imputed data, thus the number of evaluable outcome measurements at the time of interest will also be presented. There will be no formal statistical analysis of differences in baseline characteristics between treatment groups. Any clinically important imbalance between the treatment groups will be noted.

4.4 Study drug exposure, and rescue therapy

In the blinded review of the data it was observed that 13/78 patients received the maximum treatment dose of 25mg (patients with weight ≥ 100 kg). Rest of the patients complied with the dosing regimen of 0.25mg/kg. Bleeding risk increases immediately after administrating the study drug and stays increased the first 2-3 days. Rescue therapy is seldomly used. Thus, no further summaries of study drug exposure or rescue therapy will be provided.

4.5 Outcome Definitions

4.5.1 General Definitions and Derived Variables

4.5.1.1 Visual acuity of at least 0.7 logMAR 30 (±5) days after treatment (primary endpoint)

The primary endpoint is the proportion of patients with at least 0.7 logMAR visual acuity 30 (±5) days after treatment (logMAR ≤ 0.7), representing an improvement in visual acuity of at least 0.3 logMAR, equal to 15 letters/three lines on a visual acuity chart. The binary primary outcome variable “logMAR ≤ 0.7 30 days after treatment, yes/no” is defined based on the ETDRS letter score registered at the 30-day follow-up visit. ETDRS letter score can be converted to logMAR value by the following formula: logMAR = 1.7-0.02*ETDRS letter score. Thus, an ETDRS score of 50 corresponds to logMAR-value of 0.7. At 30-day follow-up visit, patients with ETDRS letter score of 50 or greater, are assigned a “logMAR ≤ 0.7” -value 1 (improvement of at least 0.3 logMAR), and patients with ETDRS letter score <50 are assigned a value 0 (no improvement). Patients who are not able to read the chart will be assigned a value 0. See point 4.6.1.4 for handling of missing values.

4.5.1.2 Visual acuity of at least 0.5 logMAR 30 (±5) and 90 (±15) days after treatment (secondary endpoint)

The primary endpoint is the proportion of patients with at least 0.5 logMAR visual acuity 30 (±5) days after treatment (logMAR ≤ 0.5), representing an improvement in visual acuity of at least 0.5 logMAR. The binary secondary outcome variables “logMAR ≤ 0.5 30 days after treatment, yes/no” and “logMAR ≤ 0.5 90 days after treatment, yes/no” are defined based on the ETDRS letter score registered at the 30/90-day follow-up visit. An ETDRS score of 60 corresponds to logMAR-value of 0.5. Thus, patients with ETDRS letter score of 60 or greater at 30/90-day follow-up visit, are assigned a “logMAR ≤ 0.5” -value 1 (improvement of at least 0.5 logMAR), and patients with ETDRS letter score <60 are assigned a value 0 (no improvement). Patients who are not able to read the chart will be assigned a value 0.

4.5.1.3 Change from baseline to day 30/90 in the best-corrected logMAR visual aquity

Visual acuity is registered at baseline using best-corrected decimal visual acuity (BVCA). Profound vision loss (patient is not able to read the chart) is graded by counting fingers (CF), handmotion (HM), light perception (LP) or no light perception (NLP). At 30-day and 90-day follow-up visits, ETDRS letter score is registered. BVCA and ETDRS are converted to logMAR-values for statistical analyses as follows:

- logMAR = -log10(BVCA)
- logMAR = 1.7-0.02*ETDRS letter score

Based on the blinded review of the data, n=13/9/10 patients were able to detect finger count, n=32/23/20 hand motion, n=22/10/4 light perception and n=7/3/3 no light perception with the affected eye at baseline/30 days/90 days, respectively. Sligthly different values for converting finger count and hand motion to logMAR have been presented, but here we have chosen values based on data from Schulze-Bonsels et al. (2006)[2] and Lange et al. (2009)[3].

- counting fingers: logMAR = 2.0
- hand motion: logMAR = 2.3

Light perception is not a visual acuity measurement, but simply detection of a stimulus. However, presenting it on a numeric scale is useful in order to quantitatively document intervention success. We chose to use logMAR=2.7 for patients with light perception, and logMAR=3.0 for patients with no light perception.

The continuous secondary endpoint “change from baseline in visual acuity at 30/90 days after treatment” will be defined as a difference between the logMAR visual acuity at baseline and at 30/90 days. Negative values of the difference reflect improvement.

4.5.1.4 EQ-5D-5L (secondary endpoint)

EQ-5D-5L is a standardised health related quality of life instrument consisting of two components, health state description and evaluation. For the description component a subject self-rates their health in five dimensions (mobility, self-care, usual activities, pain/comfort and anxiety/depression) using five level scale (no problems to unable/extreme problems). The evaluation component requires a subject to record their overall health status on a visual analogue scale (0-100). The description component index values are calculated using five levels (5L) according to the EQ-5D Norway value set produced using the standardized valuation study protocol (EQ-VT).

4.5.1.5 NIH Stroke Scale score (NIHSS)

NIH Stroke scale is a numerical scale to determine stroke severity by evaluating patient’s performance in 11 categories, such as sensory and motor ability. The total score is a simple sum of the scores of the 11 categories, ranging from 0 to 42 points, with higher numbers indicating greater severity. A score of <5 represents no stroke symptoms or a minor stroke, a score of 5 to 15 represents a moderate stroke, a score of 16 to 20 represents a moderate to severe stroke, and a score of 21 to 42 represents a severe stroke.

4.5.1.6 Modified Rankin Scale score (mRS)

The Modified Rankin Score (mRS) is a six point disability scale with possible scores ranging from 0 (no symptoms at all) to 5 (severe disability; bedridden, incontinent and aquiring constant nursing care and attention). A separate category of 6 is added for patients who die.

4.5.1.7 Acute ischemic lesions on follow-up diffusion-weighted (DWI) MRI or brain CT

The presence of acute ischemic lesions on DWI MRI or brain CT is defined based on the Alberta Stroke Program Early CT score (ASPECTS) in the core lab imaging assesment data. A 10-point quantitative score divides middle cerebral artery (MCA) into ten regions of interest and the MCA territory is allotted 10 points. A single point is subtracted for an area of early ischemic change, such as focal swelling or parenchymal hypoattenuation, for each of the defined regions. A normal CT scan receives an ASPECTS of 10 points. A score of zero indicates diffuse ischemic involvement throughout the MCA territory. Here, the safety endpoint “acute ischemic lesions on DWI MRI or brain CT” gets a value 1 (present) if patient has an ASPECTS <10, and 0 (not present) if ASPECTS = 10.

4.5.1.8 Any intracranial haemorrhage

The corelab imaging analysis follows the Heidelberg Bleeding Classification (HBC) for haemorrhagic findings. The values are stored in two variables, reflecting that the HBC allows for multiple simultaneous classifications. For each time point (Day 0 and Day 1), the data are recorded as follows:

a) HBC – haemorrhagic transformation (0, 1a, 1b, 1c, 2): where 0 indicates no haemorrhage and the values 1a to 2 represent increasing severity of haemorrhage within an infarcted area.
b) b) HBC – haemorrhage outside infarct (None, 3a, 3b, 3c, 3d): where None indicates no haemorrhage, and 3a–3d indicate haemorrhage outside the infarct.

In principle, any HBC classification other than '0' (for transformation) and 'None' (for haemorrhage outside infarct) would qualify a positive finding for safety endpoint “*any intracranial bleeding*”. However, there is a potential inconsistency due to the inclusion of both MRI and CT modalities. MRI is more sensitive to subtle haemorrhagic transformations, and HBC 1a (and occasionally 1b) may not be detectable on CT. Therefore, for the purposes of statistical analysis, HBC None (haemorrhage outside infarct), 0, 1a, and 1b (haemorrhagic transformation) are all treated as “no haemorrhage”, and 1c, 2, 3a-3d are treated as “haemorrhage”.

4.5.1.9 Symptomatic intracranial haemorrhage

Symptomatic intracranial haemorrhage is defined as presence of any intracranial haemorrhage and NIHSS detoriation of more than 4 points.

4.5.1.10 NEI VFQ-25 total score

The secondary outcome “National Eye Institute Visual Function Questionnaire” (NEI VFQ-25) is a 25-item questionnaire with a total score that is computed in three steps as follows[4]:

1. Original numeric values for each item in the questionnaire are re-coded following the scoring rules outlined in Table 5.1. All items are scored so that a high score represents better functioning. Each item is then converted to 0-100 scale as shown in Table 5.1 (lowest and highest possible scores are set at 0 and 100 points, respectively).

2. Items within each subscale are averaged together to create 12 sub-scale scores (general health, general vision, well-being/distress, ocular pain, near vision, distance vision, peripheral vision, social functioning, color vision, role limitations, driving, dependency). Table 5.2 indicates which items contribute to which sub-scale. Items that are left blank (missing) are not taken into account when calculating the scale scores. Sub-scales with at least one item answered can be used to generate a sub-scale score.
3. To calculate the overall composite score for the VFQ-25, all vision targeted sub-scale scores are averaged, excluding the general health rating question. A total score can get values between 0 and 100, 0 being the worst possible score and 100 the best possible score.

Table 5.1: Recoding of items to scale 0-100 or missing.

Item Numbers	Change original response category [a]	To recoded value of:
1,3,4,15c[b]	1	100
	2	75
	3	50
	4	25
	5	0
2	1	100
	2	80
	3	60
	4	40
	5	20
	6	0
5,6,7,8,9,10,11,12,13,14,16,16a A3,A4,A5,A6,A7,A8,A9[c]	1	100
	2	75
	3	50
	4	25
	5	0
	6	*
17,18,19,20,21,22,23,24,25, A11a,A11b,A12,A13	1	0
	2	25
	3	50
	4	75
	5	100
A1,A2	0	0
	to	to
	10	100

[a] Precoded response choices as printed in the questionnaire.

[b] Item 15c has four-response levels, but is expanded to a five-levels using item 15b.
Note: If 15b=1, then 15c should be recoded to "0"
If 15b=2, then 15c should be recoded to missing.
If 15b=3, then 15c should be recoded to missing.

[c] "A" before the item number indicates that this item is an optional item from the Appendix. If optional items are used, the NEI-VFQ developers encourage users to use <u>all</u> items for a given sub-scale. This will greatly enhance the comparability of sub-scale scores across studies.

* Response choice "6" indicates that the person does not perform the activity because of non-vision related problems. If this choice is selected, the item is coded as "missing."

Table 5.2: Averaging of items to generate VFQ-25 subscales.

Scale	Number of items	Items to be averaged (after recoding per Table 2)
General Health	1	1
General Vision	1	2
Ocular Pain	2	4, 19
Near Activities	3	5, 6, 7
Distance Activities	3	8, 9, 14
Vision Specific:		
Social Functioning	2	11, 13
Mental Health	4	3, 21, 22, 25
Role Difficulties	2	17, 18
Dependency	3	20, 23, 24
Driving	3	15c, 16, 16a
Color Vision	1	12
Peripheral Vision	1	10

4.5.2 Overview of Outcomes

Level	Outcome	Timeframe	Type	Primary publication	Follow-up publication
Primary outcome					
Primary	≤ 0.7 logMAR best-corrected visual acuity in the affected eye	D30	Dichotomous	x	
Secondary outcomes					
Secondary	≤ 0.5 logMAR best-corrected visual acuity in the affected eye	D30	Dichotomous	x	
Secondary	≤ 0.5 logMAR best-corrected visual acuity in the affected eye	D90	Dichotomous		x
Secondary	Change from baseline in logMAR visual acuity in the affected eye at 30 days	D30	Continuous	x	
Secondary	Change from baseline in logMAR visual acuity in the affected eye at 90 days	D90	Continuous		x
Secondary	Number of test points seen (of 100) on monocular Esterman perimetry	D30	Continuous	x	
Secondary	Number of test points seen (of 100) on monocular Esterman perimetry	D90	Continuous		x
Secondary	Acute ischemic lesions on follow-up diffusion-weighted (DWI) MRI or brain CT	D1	Dichotomous	x	
Secondary	National Institutes of Health Stroke Scale score	Discharge	Continuous	x	
Secondary	Modified Rankin Scale score	Discharge	Categorical	x	
Secondary	Modified Rankin Scale score	D30	Categorical	x	
Secondary	Modified Rankin Scale score	D90	Categorical		x
Secondary	NEI VFQ-25 total score	D30	Continuous	x	
Secondary	NEI VFQ-25 total score	D90	Continuous		x
Secondary	EQ-5D-5L	D30	Continuous	x	
Secondary	EQ-5D-5L	D90	Continuous		x
Secondary	Presence of ocular neovascularization in the affected eye	D30	Dichotomous	x	
Secondary	Presence of ocular neovascularization in the affected eye	D90	Dichotomous		x
Safety outcomes					
Safety	All-cause death	Discharge	Dichotomous	x	
Safety	All-cause death	D30	Dichotomous	x	
Safety	All-cause death	D90	Dichotomous		x
Safety	Stroke-related death	Discharge	Dichotomous	x	
Safety	Stroke-related death	D30	Dichotomous	x	
Safety	Stroke-related death	D90	Dichotomous		x
Safety	Any intracranial haemorrhage	D1	Dichotomous	x	
Safety	Symptomatic intracranial hemorrhage	Discharge	Dichotomous	x	
Safety	Complications (such as systemic bleeding)	D1	Dichotomous	x	

Safety	Complications (such as systemic bleeding)	Discharge	Dichotomous	x	
Safety	Complications (such as systemic bleeding)	D30	Dichotomous	x	
Safety	Complications (such as systemic bleeding)	D90	Dichotomous		x
Safety	Serious adverse events other than intracranial haemorrhage	Whole study period until D30	Dichotomous	x	
Safety	Serious adverse events other than intracranial haemorrhage	Whole study period until D90	Dichotomous		x
Safety	Any adverse events	Whole study period until D30	Dichotomous	x	
Safety	Any adverse events	Whole study period until D90	Dichotomous		x

4.6 Analysis Methods

4.6.1 Primary Outcome

4.6.1.1 Primary Analysis

The primary analysis will be performed in the FAS. The primary outcome variable is the occurrence of improvement of at least 0.3 logMAR best-corrected visual acuity (logMAR $\leq$ 0.7) at 30 day follow-up visit. The primary outcome will be analyzed using a logistic regression model with treatment group as a dichotomous covariate (Tenecteplase Placebo + ASA = 0, Tenecteplase + ASA Placebo = 1).

As there are multiple study sites with only one or two patients included, including the study site as a fixed covariate in the model is potentially problematic. For the patients at small sites, the information would only be used to estimate the site effect, possibly diluting the estimate of the treatment effect. A short simulation performed in the blinded data review and mock analyses showed that unless there is a large true site effect, there will be a small drop in power if the model when adjusting for the site as a fixed covariate. In the blinded mock analysis of the primary outcome data, site was also considered as a random effect (random intercept). However, this led to singular model fit with several different permutations of the mock treatment group allocation (close to zero random effects). Thus, in the primary analysis, study site will not be included in the model, but a robustness analysis adjusted for the study site (fixed effect) will be performed.

The primary null hypothesis is that there is no difference in the probability of visual recovery (logMAR $\leq$ 0.7) within acuity 30 days after study treatment. The alternative hypothesis is that there is a difference in the probability of visual recovery (logMAR $\leq$ 0.7) within 30 days after study treatment. The primary effect estimate will be the population-averaged (marginal) adjusted risk difference of logMAR $\leq$ 0.7 between the groups. The adjusted relative risk of sustained inactive disease comparing the two groups will also be reported. Both estimates are reported with 95% CIs and the p-value of the null-hypothesis test of no treatment difference from the logistic regression.

4.6.1.2 Summary Measures

The primary analysis will be presented with the number and percentage of patients reaching visual recovery by treatment group, the estimate of the primary estimator with 95% confidence limits, and the p-value of the primary null hypothesis. The null hypothesis will be evaluated on the 5%

significance level. As there is only one primary analysis, there will be no adjustments for multiplicity testing.

The treatment effect on the primary endpoint will be reported with the treatment effect parameter estimate, standard error and the p-value. In addition, the population-averaged (marginal) *adjusted risk difference* and population-averaged adjusted *relative risk* of visual recovery between the groups will be presented with 95% confidence interval. The CI will be calculated from the logistic regression using the delta method and implemented by R-package *marginaleffects* and function *avg_comparisons*.

The numbers needed to treat (NNT) will obtained by taking the reciprocal of the primary estimator (risk difference). An estimate of NNT will only be presented if the estimated 95% CI of the difference in probability of visual recovery excludes zero.

4.6.1.3 Assumption Checks and Alternative Analyses

As there are no continuous covariates in the logistic regression, there will be no assumption checks performed for the primary analysis.

4.6.1.4 Missing Data

For the purpose of the primary analysis, the missing values of the primary outcome will be imputed using the following rules:

- If patient missed the 30-day evaluation (for example, due to severe illness), but completed the 90-day evaluation, the 90-day log-MAR best-corrected visual acuity can be used to impute the missing 30-day value.
- If there are no measurements of logMAR after the study treatment (for example, due to death or other withdrawal), the patient is regarded as not having visual recovery at day 30 (worst case imputation)

Sensitivity analyses will be run using only complete cases, worst case imputation for all patients, and best case imputation for all patients.

4.6.1.5 Sensitivity Analyses

- Primary analysis model adjusted for study site (fixed effect)
- Missing data handling: complete case analysis
- Missing data handling: worst case imputation
- Missing data handling: best case imputation (all missing as logMAR ≤ 0.7)

As the difference between FAS and PP is only one patient, sensitivity analysis in PP will not be performed.

4.6.1.6 Subgroup Analyses

Subgroup analyses will be performed by including an interaction term between the variable in question and the dichotomous treatment variable in the model for the primary outcome. Subgroup analyses based on the following variables will be performed:

- Administration time of the study treatment

    - <3h from symptom onset
    - 3 to 4.5h from symptom onset
- Etiology of CRAO:
    - Carotid artery atherosclerosis
    - Carotid artery dissection
    - Cardioembolism due to atrial fibrillation (AF)
    - Cardioembolism due to non-AF cardiac disease
    - Small vessel disease
    - Other causative disease
    - Unknown

The results from the subgroup analysis will be presented by displaying the confidence intervals of the risk difference (active – placebo) in a forest plot.

4.6.2 Dichotomous Secondary Outcomes

All secondary analyses will be performed in the FAS. The following dichotomous secondary outcomes (yes/no) will be analyzed using a logistic regression model similar to primary outcome

- ≤ 0.5 logMAR best-corrected visual acuity in the affected eye
- Acute ischemic lesions on follow-up diffusion-weighted (DWI) MRI or brain CT
- Presence of ocular neovascularisation in the affected eye

4.6.2.1 Main Analysis

The analyses methods and estimators will be permorfed similarly to the primary analyses.

4.6.2.2 Summary Measures

The treatment effects on the secondary dichotomous endpoints will be reported as the population-averaged (marginal) *adjusted risk difference* of visual recovery/ presence of acute ischemic lesions/ presence of ocular neovascularisation between the groups will be presented with 95% confidence interval.

4.6.2.3 Missing Data

Missing data in the secondary outcome "≤ 0.5 logMAR best-corrected visual acuity in the affected eye" will be imputed similarly to primary outcome (see 4.1.6.4).

Missing data for "presence of ocular neovascularization" at 30 days will be imputed with 90-day value, if available. Missing data at 90 days will be imputed with 30-day value, if available. If both values are missing, they will be imputed with the "no presence" as the event is expected to be rare. Blinded review of the data before the primary publication showed that "no presence" value imputation would be needed only one patient, who died shortly after the treatment.

4.6.2.4 Sensitivity Analyses

No sensitivity analyses will be performed for the secondary outcomes.

4.6.2.5 Subgroup Analyses

The secondary outcome "≤ 0.5 logMAR best-corrected visual acuity in the affected eye" will be analysed in the following subgroups determined by the administration time of the study treatment:

- <3h from symptom onset
- 3 to 4.5h from symptom onset

4.6.3 Categorical Secondary Outcomes

All secondary analyses will be performed in the FAS. The categorical secondary outcome "modified Rankin scale score" has seven categories:

- 0=No symptoms at all
- 1=No significant disability despite symptoms; able to carry out all usual duties and activities
- 2=Slight disability; unable to carry out all previous activities, but able to look after own affairs without assistance
- 3=Moderate disability; requiring some help, but able to walk without assistance
- 4=Moderately severe disability; unable to walk without assistance and unable to attend to own bodily needs without assistance
- 5=Severe disability; bedridden, incontinent and aquiring constant nursing care and attention
- 6=Dead.

4.6.3.1 Main Analysis

Rankin scale score will be analyzed using an ordinal regression model with mRS as an ordinal outcome and treatment group as a covariate. Three models will be fitted separately at each time point (discharge, 30 days, 90 days).

4.6.3.2 Summary Measures

The treatment effect on mRS will be reported as the odds ratio (change in odds of being at or above a certain Rankin scale level for intervention group in comparison to the placebo group) together with 95% confidence interval.

4.6.3.3 Assumption Checks and Alternative Analyses

The proportional odds assumption will be checked using a Brant-test which assesses whether the observed deviations from the ordinal logisitic regression are larger than what would be attributed to chance alone. If the probability is greater than 0.05, then the data satisfies the proportional odds assumption. In case the proportional odds assumption is not met, the association between treatment group and mRS will be assessed with a Chi-square or Fisher's exact test, and no treatment effet measure will be reported.

4.6.3.4 Missing Data

Missing data for "Rankin scale score"will be imputed using last observation carried forward.

4.6.3.5 Sensitivity Analyses

No sensitivity analyses will be performed for categorical secondary outcomes.

4.6.3.6 Subgroup Analyses

No subgroup analyses will be performed for categorical secondary outcomes.

4.6.4 Continuous Secondary Outcomes

The following continuous secondary outcomes are measured repeatedly and will be analysed using a linear regression with treatment group as a fixed covariate:

- Number of test points seen on monocular Esterman perimetry
- Change from baseline in logMAR visual acuity at 30/90 days after treatment
- National Institutes of Health Stroke Scale score NIHSS
- NEI VFQ-25 total score
- EQ-5D-5L quality of life index score

4.6.4.1 Main Analysis

Continuous secondary outcomes will be analysed using linear regression model, with the endpoint as dependent variable, and treatment group independent variables.

The blinded review of the data showed that NIHSS score (range 0-42) had very little variation on Day 1 and at Discharge, most of patients having an NIHSS score of 0, and only two patients having values 1 or 2. At Discharge (>36h after treatment), approx. 40% of the patients had a missing NIHSS score. The observed values (0/1/2) of the NIHSS score will be tabulated and the association between treatment group and NIHSS score analysed using Fisher's exact test.

4.6.4.2 Summary Measures

The continuous secondary outcomes will be presented descriptively with mean (SD) / median (IQR) at each time point. The efficacy estimator of effect will be the marginal mean difference between the treatment groups.

4.6.4.3 Assumption Checks

Normality of the residuals will be checked visually using Q-Q-plots. In case of non-normal residuals, a non-parametric test, such as Mann-Whitney U-test will be performed to compare the mean differences in the outcome between the treatment groups.

4.6.4.4 Missing Data

- Missing data in logMAR best-corrected visual acuity at 30 days will be imputed with 90-day value, if available. Missing data in logMAR at 90 days will be imputed with 30-day value, if available. If both 30- and 90-day logMAR are missing, they will be imputed with the baseline logMAR (resulting in no change).
- Missing values in NIHSS score will be imputed with last observation carried forward.
- Missing values in test points seen on monocular Esterman perimetry at 30 days will be imputed with 90-day value, if available. Missing data at 90 days will be imputed with 30-day value, if available. If both 30- and 90-day values are missing, they will be imputed with the worst score value of 0.
- Missing values for NEI VFQ-25 at 30 days will be imputed with 90-day value, if available. Missing data at 90 days will be imputed with 30-day value, if available. If both values are missing, they will be imputed with the worst score value of 0. Blinded review of the data before the primary publication showed that worst value imputation would be needed only one patient, who died shortly after the treatment.

- Missing values for EQ-5D-5L at 30-days will be imputed with 90-day value, if available. Missing data at 90 days will be imputed with 30-day value, if available. If both values are missing, they will be imputed with the lowest index score value of -0.59. Blinded review of the data before the primary publication showed that worst value imputation would be needed only one patient, who died shortly after the treatment.

4.6.4.5 Sensitivity Analyses
No sensitivity analyses will be performed for continuous secondary outcomes.

4.6.4.6 Subgroup Analyses
No subgroup analyses will be performed for continuous secondary outcomes.

5 Safety Analyses

General safety evaluations will be descriptive and based on the incidence, intensity, and type of adverse event (AE). Safety variables will be tabulated and presented for all patients in the safety analysis set. Study group will go through all registered AEs in a blinded review, and mark the ones that are considered stroke-related death and complications.

Safety endpoints:

- All-cause and stroke-related death at discharge, 30 (±5) and 90 (±15).
- Proportion of patients with any intracranial haemorrhage at 24 hours.
- Proportions of patients with symptomatic intracranial haemorrhage until discharge.
- Proportion of patients with complications such as systemic bleeding at 24 hours, discharge and 30 (±5) days.
- Other serious adverse events.
- Occurrence of adverse events.

5.1 Adverse Events
An AE is any untoward medical occurrence in a patient administered a pharmaceutical product and which does not necessarily have a causal relationship with this treatment.

An AE can therefore be any unfavorable and unintended sign (including an abnormal laboratory finding), symptom, or disease temporally associated with the use of a medicinal (investigational) product, whether or not related to the medicinal (investigational) product.

The term AE is used to include both serious (SAEs) and non-serious AEs.

A Serious Adverse Event (SAE) is defined as a medical occurrence that at any dose:

- Results in death
- Is immediately life-threatening
- Requires in-patient hospitalization or prolongation of existing hospitalization
- Results in persistent or significant disability or incapacity

- Is a congenital abnormality or birth defect
- Is an important medical event that may jeopardize the subject or may require medical intervention to prevent one of the outcomes listed above.

A Suspected Unexpected Serious Adverse Reaction (SUSAR) is a SAE that is unexpected. Expected Adverse Events (EAE) include all events listed in the SmPC; EAEs are not reported.

For each patient, the standard time period for collecting and recording AE and SAEs will begin at start of study treatment and continue until the end of the trial. All events will be categorized and graded according to MedDRA (Medical Dictionary for Regulatory Activities).

For tabulations, only treatment emerging adverse events (TEAEs) will be presented; TEAEs are defined as AEs with a start date on or after date of first randomized treatment until 7 days following the last dose of the study drug.

We will present number and % of subjects by treatment group with:

- Any TEAE
- 1, 2 or > 3 TEAEs
- Any treatment emerging serious AE (TESAE)
- TEAEs leading to study drug withdrawal
- Mild, moderate and severe TEAE
- TEAE unrelated, unlikely, possible, probably and definite related to study drug
- Ongoing and resolved TEAE

The number of events and number (%) of subjects with TEAE by system organ class (SOC) and preferred term (PT) will be summarized by treatment group, 1) overall, 2) for severe TEAEs and 3) for TEAEs leading to study discontinuation.

5.2 Clinical Laboratory Parameters

Safety clinical laboratory parameters were collected and assessed, but only used to identify adverse events. No clinical laboratory parameters will be reported.

5.3 Vital Signs

Changes in vital signs were collected and assessed, but only used to identify adverse events. No vital signs will be reported.

6 Statistical Software

All statistical analyses will be done using R, v4.5.0 (www.r-project.org).

7 References

7.1 Literature References

1 Gamble, C. *et al.* Guidelines for the Content of Statistical Analysis Plans in Clinical Trials. *JAMA* **318**, 2337-2343 (2017). https://doi.org/10.1001/jama.2017.18556

2 Schulze-Bonsel K, et. al. Visual acuities “Hand Motion” and “Counting Fingers” can be quantified using the Freiburg Visual Acuity Test. Invest Ophthalmol Vis Sci **47**, 1236-1240 (2006).

3 Lange C, et. al. Resolving the clinical acuity categories “hand motion” and “counting fingers” using the Freiburg Visual Acuity Test (FrACT). Graefe’s Arch Clin Exp Ophthalmol **247**, 137–142 (2009).

4 Mangione, C. M. *et al.* Development of the 25-item National Eye Institute Visual Function Questionnaire. *Arch Ophthalmol* **119**, 1050-1058 (2001).

7.2 Reference to other Standard Operating Procedures or Documents

Data tables and figures will be configured according to publication requirements. Data listings will be provided as needed.